**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**

**Department of Artificial Intelligence and Machine Learning**

# ASSESSMENT 2 – Scenario-Based Questions (5 QUESTIONS)

| | | | |
|---|---|---|---|
| **Course Code:** | ITA05 | **Course Title:** | **Computer Vision** |
| **CO Assessed:** | CO2 | **Assessment:** | Assessment 2: Scenario-Based Questions |
| **Weightage:** | 35% of CIA | **Total Marks:** | (5 Questions × 10 Marks) |
| **CO2 Statement:** | ***Apply image enhancement, filtering, and segmentation techniques to analyze images, detect edges and boundaries, and solve real-world Computer Vision problems.*** | | |
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| **Section / Batch:** | B TECH AIML | **Date:** | 28/10/26 |

**Code of Conduct**

*I* SWATHI K (192525477) *(NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: SWATHI K

**QUESTIONS: 5** **Total Marks: 50**

## Annexure A

## Scenario-Based Qns

### Q1. Image Enhancement Using Grey-Level Transformations

A hospital uses a digital imaging system to capture X-ray images, but some images appear too dark to identify important anatomical structures. The image processing team decides to apply grey-level transformation techniques to improve image visibility. Apply a suitable transformation to enhance the brightness and contrast of the image. Calculate the transformed intensity values for selected pixels using the chosen transformation. Explain how the transformation improves the visibility of important features in the X-ray image.

### Q2. Histogram Processing and Spatial Filtering

A security surveillance system captures images at night, resulting in poor contrast and unwanted noise. The system must enhance the images to identify people and objects more clearly. Apply histogram equalization to improve the contrast of the captured image. Use an appropriate spatial filtering technique to reduce noise while preserving important details. Compare the effects of both techniques and explain how they improve the quality of the surveillance images.

### Q3. Image Smoothing and Sharpening Using Frequency Domain Filtering

A satellite imaging system captures land images containing noise and blurred boundaries between different geographical regions. The image processing team decides to use frequency domain filtering to improve image quality. Apply a low-pass filter to reduce high-frequency noise and smooth the image. Use a high-pass filter to enhance edges and sharpen important geographical features. Explain the differences between the two filters and determine how each technique contributes to image interpretation.

### Q4. Image Segmentation Using Edge Detection and Thresholding

An automated manufacturing system must inspect metal components to identify cracks and surface defects. The captured images contain variations in brightness, making defect detection difficult. Apply an appropriate edge detection technique to identify the boundaries of possible cracks. Use thresholding to separate defective regions from the background and construct a binary image representing the detected defects. Explain how the selected techniques help the inspection system identify defective components.

### Q5. Region-Based Segmentation and Boundary Detection

An agricultural monitoring system uses aerial images to identify crop fields, water bodies, and surrounding land regions. The captured images contain regions with similar intensity values and some discontinuous boundaries. Apply region-based segmentation to group neighbouring pixels with similar characteristics. Use edge linking and boundary detection techniques to connect broken edges and identify the boundaries of different regions. Explain how these techniques help distinguish agricultural fields and improve the accuracy of land-region identification.

**Rubrics for Calculation**

| Criteria | Weightage | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| **Technical Content Accuracy** | 30 | Highly accurate, indepth, and insightful technical explanation | Technically correct content with adequate depth and clarity | Basic concepts presented with limited accuracy and depth | Content contains major technical errors; concepts poorly understood |
| **Problem Identification** | 25 | Problem rigorously analyzed using appropriate and advanced methods | Problem clearly defined with a logical engineering approach | Problem identified but analysis or methodology is weak | Problem statement unclear or incorrect; no logical approach |
| **Use of Tools** | 20 | Advanced tools, well-analyzed data, and highly effective visuals | Appropriate tools and data used effectively with clear visuals | Limited use of tools and basic data interpretation | Minimal or incorrect use of tools and data; poor visuals |
| **Communication** | 15 | Highly engaging, confident, and audience-focused delivery | Clear, organized, and professional communication | Understandable but lacks clarity, structure, or confidence | Poor organization, unclear speech, ineffective slides |
| **Professionalism** | 10 | Exemplary professionalism, leadership, and ethical awareness | Professional conduct with effective team collaboration | Basic professionalism; uneven team participation | Lacks professionalism; minimal contribution or ethical awareness |

| Faculty In-charge | Course Coordinator | Program Director |
|---|---|---|
| Signature: _____ | Signature: _____ | Signature: _____ |
| Date: _____ | Date: _____ | Date: _____ |

# Q1. Image Enhancement Using Grey-Level Transformations

## 1.1 Problem Identification

A hospital X-ray system produces images that are under-exposed: almost all pixels lie in a narrow, dark band of the grey scale. In our test image the grey levels occupy only 5 to 70 out of the 0–255 range (mean = 22.7, standard deviation = 11.4). Because the human eye cannot separate such closely spaced dark levels, ribs, lung fields and the spine appear merged with the background. The objective is to expand the dark grey levels so that the anatomical structures become distinguishable, without needing to re-expose the patient to radiation.

This is a point-processing (spatial domain) problem. The output at a pixel depends only on the input value at that pixel: $s = T(r)$, where $r$ is the input grey level, $s$ is the output grey level and $T$ is the transformation function. For an 8-bit image $L = 256$, so $0 \le r, s \le 255$.

## 1.2 Selection of Suitable Transformations

Three grey-level transformations are suitable for dark images. All of them give a steep slope for low input values, which spreads the dark levels over a wider output range.

- **Log transformation:** $s = c \cdot \ln(1 + r)$, with $c = 255 / \ln(256) = 45.99$. It strongly expands dark values and compresses bright ones.
- **Power-law (gamma) transformation:** $s = 255 \cdot (r / 255)^{\gamma}$. For $\gamma < 1$ (we choose $\gamma = 0.4$) it behaves like the log function but gives a tunable amount of correction through the single parameter $\gamma$.
- **Contrast stretching:** a piecewise-linear function $s = 255\ (r - r_{min}) / (r_{max} - r_{min})$, which maps the occupied range $[r_{min}, r_{max}]$ to the full range $[0, 255]$.

**Chosen transformation:** the power-law transformation with $\gamma = 0.4$ is applied as the main enhancement because it brightens dark X-ray regions while keeping the grey-level ordering natural, and its parameter can be tuned by the radiologist. Log and contrast-stretching results are computed for comparison.

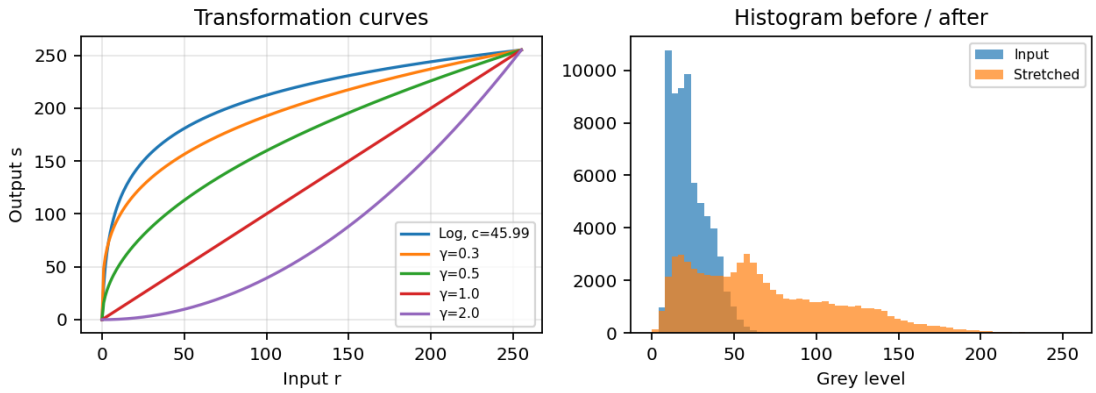



*Figure 1.1: Transformation curves (left) and histogram of input and stretched image (right)*

## 1.3 Calculation of Transformed Intensity Values

Five representative pixels from the dark X-ray are selected: r = 10 (background), 20 (soft tissue), 40 (lung), 70 (bone) and 100. The calculations are shown below.

**(a) Log transformation** (c = 45.99):

$$s = 45.99 \times ln(1 + r)$$

For r = 10: s = 45.99 × ln 11 = 45.99 × 2.3979 = 110.3 ≈ 110.

For r = 40: s = 45.99 × ln 41 = 45.99 × 3.7136 = 170.8 ≈ 171.

For r = 70: s = 45.99 × ln 71 = 45.99 × 4.2627 = 196.0 ≈ 196.

**(b) Power-law transformation** (γ = 0.4, c = 1 on normalised values):

$$s = 255 \times (r / 255)^{0.4}$$

(10/255) = 0.0392; 0.0392^0.4 = e^(0.4 × ln 0.0392) = e^(−1.295) = 0.2738;

s = 255 × 0.2738 = 69.8 ≈ 70. For r = 40: (0.1569)^0.4 = e^(−0.741) = 0.4766;

s = 121.5 ≈ 122. For r = 70: (0.2745)^0.4 = e^(−0.517) = 0.5963; s = 152.0 ≈ 152.

**(c) Contrast stretching** (r_min = 0, r_max = 70): s = (255 / 70) × r = 3.643 r. For r = 10: s = 36.4 ≈ 36; for r = 40: s = 145.7 ≈ 146; for r = 70: s = 255.

| Input r | Log s | Power-law (γ=0.4) s | Stretch s | Power-law γ=2.0 (wrong choice) s |
|---|---|---|---|---|
| 10 | 110 | 70 | 36 | 0.4 |
| 20 | 140 | 92 | 73 | 1.6 |
| 40 | 171 | 122 | 146 | 6.3 |
| 70 | 196 | 152 | 255 | 19.2 |
| 100 | 212 | 175 | — | 39.2 |

*Table 1.1: Transformed intensity values of the selected pixels*

The last column shows that choosing $\gamma > 1$ would make the image even darker (r = 70 maps to only 19), confirming that $\gamma < 1$ is the right choice for under-exposed images.

## 1.4 Experimental Result

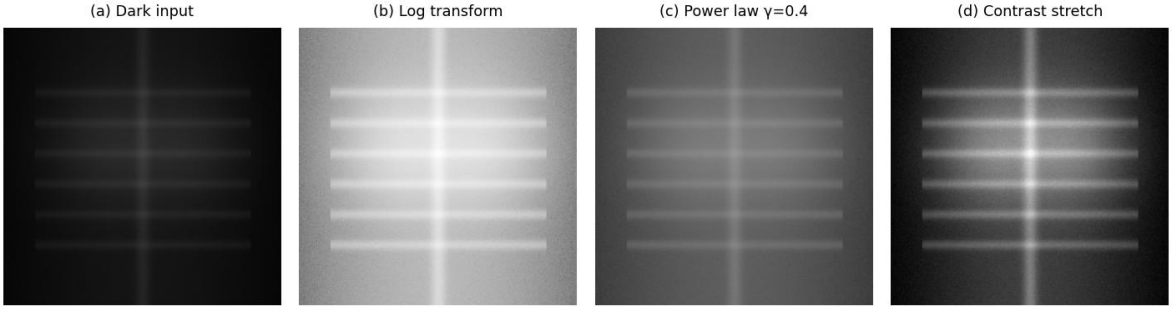



*Figure 1.2: (a) dark X-ray input, (b) log, (c) power-law γ = 0.4, (d) contrast stretching*

| Image | Mean grey level | Std. deviation (contrast) | Min | Max |
|---|---|---|---|---|
| Input (dark) | 22.7 | 11.4 | 5 | 70 |
| Log transform | 182.6 | 29.0 | 107 | 255 |
| Power-law γ = 0.4 | 94.1 | 19.1 | 53 | 152 |
| Contrast stretching | 69.5 | 44.6 | 0 | 255 |

*Table 1.2: Measured statistics of the images*

## 1.5 How the Transformation Improves Visibility

- **Brightness:** the mean grey level rises from 22.7 to 94.1 (power-law) so structures fall in the range where the eye is most sensitive.

- **Contrast:** the standard deviation rises from 11.4 to 19.1 (power-law) and 44.6 (stretching). Ribs that differed from soft tissue by only 5 to 8 grey levels now differ by about 15 to 25 levels, and become visible.
- **Dynamic-range usage:** the transformation spreads the crowded dark levels over a wide output range, so the histogram is no longer squeezed at the left end (Figure 1.1, right).
- **Trade-off:** the log transform gives the brightest result but over-brightens (mean 182.6) and compresses the bright bone detail. The power-law with $\gamma = 0.4$ gives a more balanced image, and contrast stretching gives the largest contrast. In practice the radiologist can adjust $\gamma$ until the region of interest looks best.
- **Limitation:** point transformations also amplify noise in the dark region, because noise values are mapped by the same curve. A smoothing step may follow if the noise becomes objectionable.

## 1.6 Conclusion

The power-law transformation s = 255 (r/255)^0.4 lifted the dark X-ray into a usable grey range, improving both brightness and contrast. The calculated values (10 → 70, 40 → 122, 70 → 152) match the expected behaviour, and the experiment confirms that $\gamma < 1$ or log mapping is the correct class of transformation for under-exposed medical images.

## Q2. Histogram Processing and Spatial Filtering

### 2.1 Problem Identification

A night surveillance camera works with very little light. The captured frames have two defects: (i) poor contrast: nearly all pixels are concentrated in a narrow dark band, so people and objects are not distinguishable from the background; and (ii) noise: sensor noise at high gain, appearing as random bright and dark (salt-and-pepper) pixels. In the test image the background-to-person difference is only about 7 grey levels, and the noise standard deviation in a flat area is 24.1.

Two different tools are needed because the problems are different: contrast is a global histogram property fixed by histogram equalization (HE), while noise is a local neighbourhood property fixed by spatial filtering. The surveillance system applies both in a pipeline.

### 2.2 Histogram Equalization (HE)

HE maps the grey levels so that the output histogram is approximately uniform. For a discrete image with n pixels, L grey levels and n_k pixels at level r_k:

$$s_k = T(r_k) = (L - 1) \times \Sigma\,(n_j / n),\ j = 0 \ldots k \;=\; (L - 1) \times CDF(r_k)$$

**Worked example** (3-bit image, L = 8, 64 × 64 = 4096 pixels, typical of a dark night frame):

| r_k | n_k | p(r_k) = n_k/4096 | CDF | s = 7 × CDF | Rounded s |
|---|---|---|---|---|---|
| 0 | 790 | 0.19 | 0.19 | 1.35 | 1 |
| 1 | 1023 | 0.25 | 0.44 | 3.10 | 3 |
| 2 | 850 | 0.21 | 0.65 | 4.55 | 5 |
| 3 | 656 | 0.16 | 0.81 | 5.67 | 6 |
| 4 | 329 | 0.08 | 0.89 | 6.23 | 6 |
| 5 | 245 | 0.06 | 0.95 | 6.65 | 7 |
| 6 | 122 | 0.03 | 0.98 | 6.86 | 7 |
| 7 | 81 | 0.02 | 1.00 | 7.00 | 7 |

*Table 2.1: Histogram equalization calculation*

Mapping: 0→1, 1→3, 2→5, 3→6, 4→6, 5→7, 6→7, 7→7. The new histogram is: level 1 = 790, level 3 = 1023, level 5 = 850, level 6 = 656 + 329 = 985, level 7 = 245 + 122 + 81 = 448. The output levels are now spread over 1 to 7 rather

than clustered at 0 to 2, so contrast improves. Note that the result is only approximately flat, because discrete levels cannot be split.

## 2.3 Spatial Filtering for Noise Reduction

A 3 × 3 median filter is used. It replaces each pixel by the median of its neighbourhood. Unlike the mean (box) filter, it does not average the outlier into the result; it simply discards it. It is therefore the appropriate filter for impulse (salt-and-pepper) noise, and it preserves edges because the median is always an actual neighbourhood value.

**Worked example:** a 3 × 3 window containing a salt-noise pixel (255) at the centre:

| | | |
|---|---|---|
| 52 | 55 | 61 |
| 49 | 255 | 58 |
| 50 | 54 | 57 |

Sorted values: 49, 50, 52, 54, 55, 57, 58, 61, 255, so the median = 55. The mean filter would give (52 + 55 + 61 + 49 + 255 + 58 + 50 + 54 + 57) / 9 = 691 / 9 = 76.8, a visibly wrong value that also blurs the neighbours. The median result (55) agrees with the surrounding scene.

## 2.4 Experimental Results

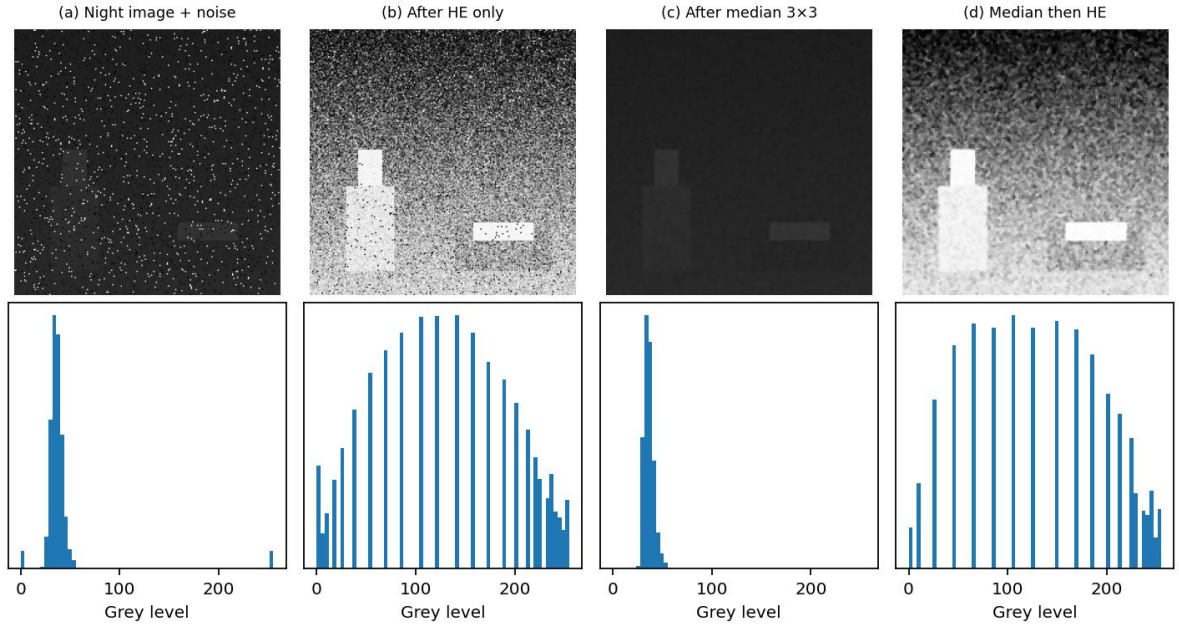



*Figure 2.1: Top row: noisy night image, HE only, median only, median then HE. Bottom row: their histograms*

| Image | Noise std in flat area | Person minus background (grey levels) | Image std (contrast) |
|---|---|---|---|
| (a) Noisy night image | 24.1 | 7.0 | 31.3 |
| (b) HE only | 45.4 | 75.8 | 74.4 |
| (c) Median 3×3 only | 1.3 | 7.6 | 4.8 |
| (d) Median, then HE | 22.0 | 79.1 | 72.1 |

*Table 2.2: Measured effect of the two techniques*

## 2.5 Comparison of the Two Techniques

| Aspect | Histogram equalization | Median spatial filter |
|---|---|---|
| Domain | Global, uses the whole histogram | Local, 3 × 3 neighbourhood |
| Main aim | Increase contrast | Remove impulse noise |
| Effect on noise | Amplifies noise (flat-area std 24.1 → 45.4) | Removes noise (24.1 → 1.3) |
| Effect on contrast | Large gain (person–background 7.0 → 75.8) | None (7.0 → 7.6) |
| Edge handling | Edges stay but noise becomes visible | Edges preserved |
| Weakness | Over-enhancement of noise and of background | Fine thin lines may be removed by large windows |

*Table 2.3: Comparison*

The results show the two techniques are complementary. HE alone makes the person visible but the noise is doubled. Median filtering alone removes the noise but the image is still dark and flat. When the median filter is applied first and HE second (image d), the noise std is 22.0, the person-to-background difference is 79.1 grey levels, and the person and vehicle are clearly separated from the scene. Filtering must come before equalization, otherwise HE stretches the noise values across the full grey range and they become hard to remove.

## 2.6 Conclusion

Histogram equalization improved the global contrast of the night images by more than 10 times, and the median filter removed impulse noise while preserving the edges of people and vehicles. Using median filtering followed by histogram equalization gives the most interpretable surveillance image.

## Q3. Image Smoothing and Sharpening Using Frequency Domain Filtering

### 3.1 Problem Identification

A satellite image of land cover contains (i) random noise from the sensor and atmosphere, and (ii) blurred boundaries between regions such as water, forest and farmland. Noise lives mostly in the high-frequency part of the spectrum (rapid pixel-to-pixel variation), and so do true edges. Smooth regions lie in the low-frequency part. Therefore the Fourier transform lets us handle the two goals separately: a low-pass filter (LPF) suppresses noise by attenuating high frequencies, and a high-pass filter (HPF) enhances edges by attenuating the low frequencies.

### 3.2 Frequency Domain Filtering Procedure

1. Multiply the input image f(x, y) by (−1)^(x+y) to centre the spectrum (or use fftshift).
2. Compute the 2-D DFT: F(u, v).
3. Multiply by the filter transfer function H(u, v): G(u, v) = H(u, v) · F(u, v).
4. Compute the inverse DFT and take the real part: g(x, y) = Re{ IDFT[G(u, v)] }.

The distance from the centre of the spectrum is $D(u, v) = \sqrt{[(u - M/2)^2 + (v - N/2)^2]}$, with M = N = 256 in our test image, and D0 is the cut-off frequency.

### 3.3 Low-Pass Filter (Smoothing)

Three standard LPFs exist. Their transfer functions are:

- **Ideal LPF:** H = 1 if D ≤ D0, otherwise 0. It has a sharp cut-off and causes ringing (ripple artefacts) near edges, because its spatial counterpart is a sinc function.
- **Butterworth LPF:** H = 1 / [1 + (D/D0)^(2n)]. The transition is gradual, so ringing is small for low order n (n = 2).
- **Gaussian LPF:** $H = \exp[-D^2 / (2\,D0^2)]$. It has no ringing at all, because the Gaussian is its own Fourier pair. It is chosen here (D0 = 30).

**Calculation example:** for the Gaussian LPF with D0 = 30, a frequency point (u, v) = (140, 150) with centre (128, 128) has $D^2 = 12^2 + 22^2 = 628$, D = 25.06. Then

H = exp(−628 / 1800) = exp(−0.349) = 0.705, so this component is kept at 70.5% of its amplitude. The table gives the three filters at several distances.

| D | Ideal LPF (D0 = 30) | Butterworth LPF (n = 2) | Gaussian LPF | Gaussian HPF = 1 − LPF |
|---|---|---|---|---|
| 0 | 1 | 1.000 | 1.000 | 0.000 |
| 15 | 1 | 0.941 | 0.882 | 0.118 |
| 30 | 1 | 0.500 | 0.607 | 0.393 |
| 60 | 0 | 0.059 | 0.135 | 0.865 |
| 90 | 0 | 0.012 | 0.011 | 0.989 |

*Table 3.1: Transfer function values for D0 = 30*

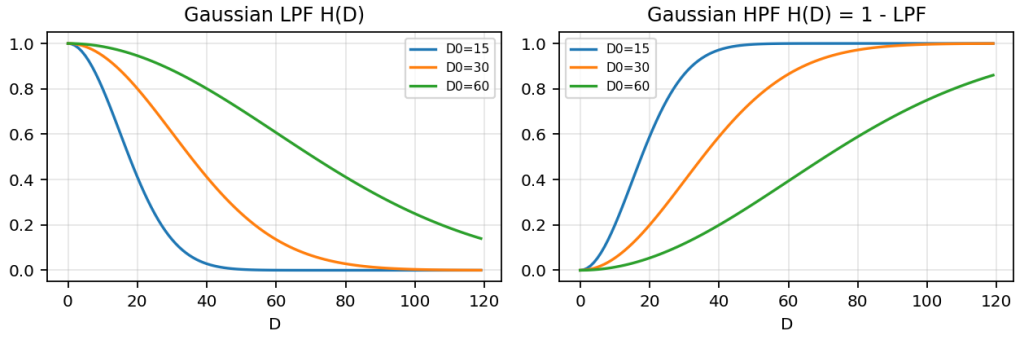



*Figure 3.1: Gaussian low-pass (left) and high-pass (right) transfer functions for different cut-offs D0*

## 3.4 High-Pass Filter (Sharpening)

The HPF is obtained as H_hp(u, v) = 1 − H_lp(u, v).

For the Gaussian: H_hp = 1 − exp[−D² / (2 D0²)].

It blocks the DC and low-frequency components (H = 0 at D = 0, from Table 3.1) and passes the edges (H = 0.865 at D = 60).

The HPF output alone shows only edges and noise on a black background (zero-mean), so for sharpening the result is added back to the original (unsharp masking / high-frequency emphasis):

$$g_{sharp}(x, y) = f(x, y) + k \cdot g_{hp}(x, y), \quad k = 1$$

The equivalent frequency-domain form is the high-frequency-emphasis filter H_hfe = a + b · H_hp with a ≥ 0 and b > a, which keeps the overall image tone while boosting the edges.

## 3.5 Experimental Results

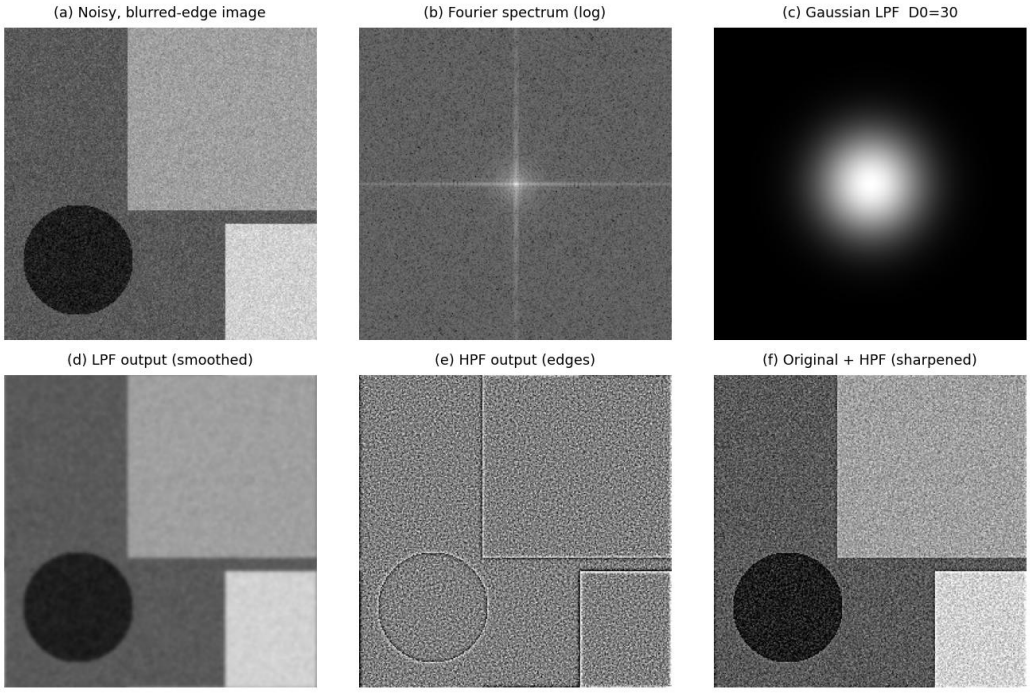



*Figure 3.2: Input, spectrum, filter, LPF output, HPF output and sharpened image*

| Image | Noise std in flat area | Step height across region boundary |
|---|---|---|
| Original (noisy) | 13.9 | 69.6 |
| After Gaussian LPF (D0 = 30) | 2.9 | 69.8 |
| Original + HPF (sharpened) | 26.8 | 69.5 |

*Table 3.2: Measured results*

## 3.6 Differences Between the Two Filters

| Property | Low-pass filter | High-pass filter |
|---|---|---|
| Frequencies passed | Low (smooth areas, overall brightness) | High (edges, fine detail, noise) |
| Visual effect | Blurring / smoothing | Sharpening / edge enhancement |
| H(0, 0) (DC term) | 1 (average brightness kept) | 0 (average removed) |
| Effect on noise | Reduces it (std 13.9 → 2.9) | Amplifies it (std 13.9 → 26.8) |
| Effect on edges | Makes them wider (blurs) | Makes them crisp |
| Typical use in satellite imaging | Denoising, region-level classification | Boundary detection, road and river tracing |

*Table 3.3: LPF vs HPF*

## 3.7 Contribution to Image Interpretation

- **LPF:** the noise standard deviation fell by about 79% (13.9 → 2.9), so the land-cover regions (water, farmland, forest) become uniform patches. This helps classification and also makes automatic thresholding reliable. The cost is that boundaries are blurred; the edge step of about 70 grey levels is preserved but spread over more pixels.
- **HPF:** the output highlights the circular water body and the straight boundaries between regions, which allows analysts to delineate them. Since it also amplifies noise (13.9 → 26.8), HPF should be applied after or together with a mild low-pass filter. A band-pass or difference-of-Gaussians filter (LPF with D0 = 60 minus LPF with D0 = 15) is the usual combined solution.
- **Choice of D0:** a small D0 smooths strongly and loses boundaries; a large D0 keeps noise. D0 = 30 (about 12% of the image size) was a good compromise here.

## 3.8 Conclusion

Frequency-domain filtering separated the satellite image into smooth content and detail. The Gaussian low-pass filter removed most of the noise without ringing, while the high-pass filter exposed and sharpened the geographic boundaries. A pipeline of low-pass denoising followed by high-frequency-emphasis sharpening gives the best interpretability.

## Q4. Image Segmentation Using Edge Detection and Thresholding

### 4.1 Problem Identification

An automatic inspection station photographs metal parts and must flag cracks and surface defects. A crack is a thin, dark line, and a pit is a small dark blob. Two difficulties are present: (i) the defects have low contrast and are thin, and (ii) brightness varies across the part (uneven lighting and reflectance). In our test image the background grey level changes from about 100 to about 180 across the part, whereas the crack is only about 70 grey levels darker than its local background. A single global threshold is therefore expected to fail.

The solution has two stages: edge detection to find where the grey level changes sharply (the crack boundaries), followed by thresholding to separate defect pixels from background and form a binary image that the system can count and measure.

### 4.2 Edge Detection: Sobel Operator

The image is first smoothed with a 5 × 5 Gaussian (σ = 1.2) to suppress noise, since derivatives amplify noise. The Sobel masks then approximate the horizontal and vertical derivatives:

| | | |
|---|---|---|
| −1 | 0 | +1 |
| −2 | 0 | +2 |
| −1 | 0 | +1 |

*Sobel Gx mask (Gy is its transpose)*

$$G_x = (z3 + 2z6 + z9) - (z1 + 2z4 + z7), \quad G_y = (z7 + 2z8 + z9) - (z1 + 2z2 + z3)$$

$$M(x, y) = \sqrt{(G_x^2 + G_y^2)}, \quad \theta = \tan^{-1}(G_y / G_x)$$

Worked example A: window on a crack edge (bright metal 150 on the left, dark crack 60 on the right):

| | | |
|---|---|---|
| 150 | 150 | 60 |
| 150 | 150 | 60 |
| 150 | 150 | 60 |

G_x = (60 + 2×60 + 60) − (150 + 2×150 + 150) = 240 − 600 = −360;

G_y = (150 + 300 + 60) − (150 + 300 + 60) = 0.

Magnitude M = 360, direction 180° (gradient horizontal, so the edge is vertical). Since 360 > T = 127, the pixel is an edge pixel.

Worked example B: window on a plain metal area (noise only):

| | | |
|---|---|---|
| 148 | 151 | 150 |
| 152 | 149 | 150 |
| 150 | 148 | 151 |

$G_x = (150 + 300 + 151) - (148 + 304 + 150) = 601 - 602 = -1$; $G_y = (150 + 296 + 151) - (148 + 302 + 150) = 597 - 600 = -3$. $M = \sqrt{(1 + 9)} = 3.16$, far below 127, so it is not an edge. This shows how the edge detector responds strongly at real boundaries and weakly on noise.

## 4.3 Thresholding to Build the Binary Image

Thresholding converts the grey image into binary: $g(x, y) = 1$ (defect) if $f(x, y) < T$, otherwise 0 (background).

- **Global (Otsu) threshold:** Otsu's method chooses T to maximise the between-class variance. For the blurred image T = 140. It works only when illumination is uniform.
- **Local / adaptive threshold:** because brightness varies, the local background B(x, y) is estimated with a large Gaussian blur ($\sigma = 20$) and a pixel is a defect if $B(x, y) - f(x, y) > 28$. This removes the illumination gradient before thresholding.

**Worked example of thresholding** on a 4 × 4 block with T = 100 (grey values on the left, resulting binary image on the right; dark pixels < 100 become defect = 1):

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 142 | 138 | 85 | 140 | 0 | 0 | 1 | 0 |
| 139 | 72 | 66 | 141 | 0 | 1 | 1 | 0 |
| 141 | 135 | 70 | 138 | 0 | 0 | 1 | 0 |
| 140 | 143 | 137 | 139 | 0 | 0 | 0 | 0 |

The binary image contains 4 defect pixels connected as a short curved line (a crack segment), which can be labelled by connected-component analysis and measured (length, area).

## 4.4 Experimental Results

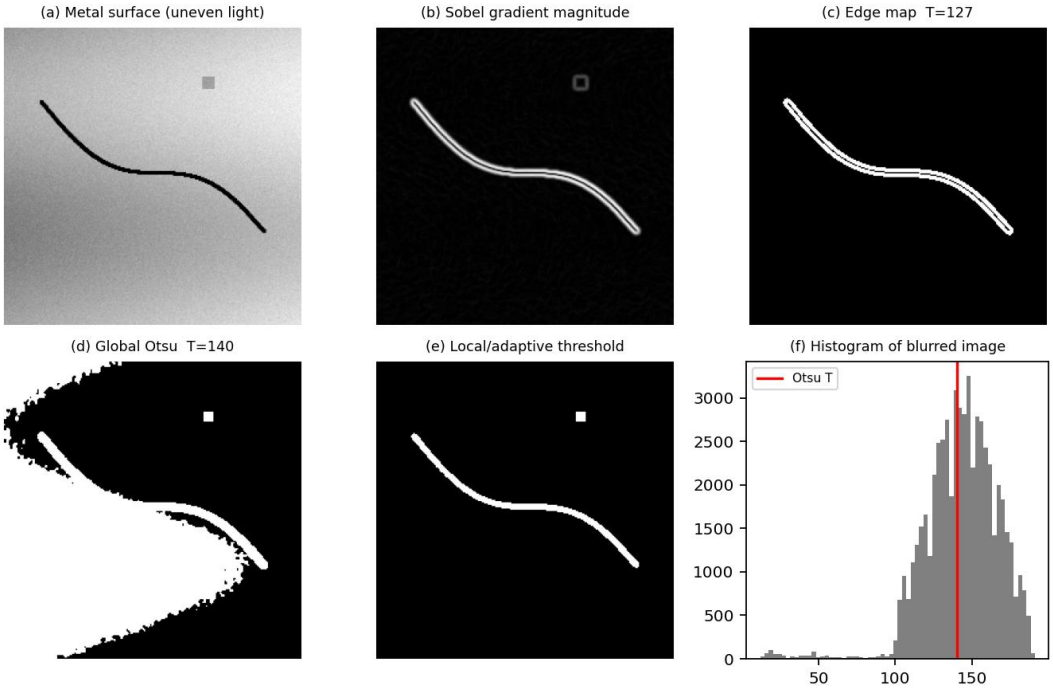



*Figure 4.1: Metal image with uneven illumination, Sobel magnitude, edge map, global Otsu result, local threshold result and histogram*

| Method | Detected pixels | True defect pixels found | False positives | Precision | Recall |
|---|---|---|---|---|---|
| Sobel edge map (T = 127) | 1408 | 613 | 795 | 0.44 | 0.67 |
| Global Otsu (T = 140) | 24157 | 882 | 23275 | 0.04 | 0.96 |
| Local adaptive threshold | 1167 | 882 | 285 | 0.76 | 0.96 |

*Table 4.1: Detection performance against the known crack and pit (918 true defect pixels)*

## 4.5 How the Techniques Help the Inspection System

- **Edge detection** gives the outline of the crack. The Sobel map traces the crack along its full length (Figure 4.1b, c) and the edge strength is independent of absolute brightness, because it depends on the difference between neighbours. However, it only gives the two borders of the crack and responds to the pit as a ring, and the recall is only 0.67 at this threshold.

- **Global thresholding fails** under uneven lighting: the dark lower-left of the part is wrongly classified as defect, giving 23275 false pixels (precision 0.04). This is the reason why the problem statement stresses brightness variation.
- **Local thresholding** corrects for the lighting gradient. It detects 882 of 918 true defect pixels (recall 0.96) with precision 0.76, and the binary image has exactly two connected components: the long crack and the small pit.
- **Decision rule:** after labelling, the system measures each component (area, length, elongation). A long thin component marks a crack, and a compact one marks a pit; if any component exceeds the tolerance, the part is rejected. A binary image makes this measurement simple and fast enough for a production line.

## 4.6 Conclusion

Sobel edge detection located the crack boundaries and local adaptive thresholding produced a clean binary defect image despite uneven brightness. Together, edge information and adaptive thresholding give reliable automatic detection of defective components (recall 96%, precision 76% in this test), while a plain global threshold is unsuitable.

## Q5. Region-Based Segmentation and Boundary Detection

### 5.1 Problem Identification

An agricultural monitoring system uses aerial images to separate crop fields, water bodies and surrounding land. The difficulty is that neighbouring regions can have similar intensity values, and that boundaries are discontinuous: edge detectors leave gaps because the intensity change is weak or noisy at some places. A pure edge-based method then produces open contours that do not enclose any region, and a pure intensity threshold cannot separate regions with overlapping grey levels. So a combination of region-based segmentation (uses similarity inside a region) and edge linking / boundary detection (uses discontinuity at the border) is required.

### 5.2 Region-Based Segmentation: Region Growing

Region growing starts from seed pixels and adds neighbouring pixels that satisfy a similarity predicate P. The algorithm is:

1. Select a seed pixel for each region (for example from the centre of a field).
2. Examine the 8-connected neighbours of the region. Add a neighbour if $|f(\text{neighbour}) - \text{seed value}| \leq T$.
3. Repeat until no more pixels can be added; then start the next seed.
4. Check that every pixel is assigned to exactly one region (completeness and disjointness): $\cup R_i = R$, $R_i \cap R_j = \emptyset$.

**Worked example:** a 5 × 5 aerial patch with threshold T = 8:

| | | | | |
|---|---|---|---|---|
| 10 | 11 | 12 | 52 | 54 |
| 11 | 12 | 13 | 53 | 55 |
| 12 | 13 | 14 | 51 | 56 |
| 60 | 62 | 15 | 50 | 57 |
| 61 | 63 | 64 | 58 | 59 |

- **Region R1** (seed = 10, accept 2 to 18): the top-left block (nine pixels, 10 to 14) plus the pixel 15 in row 4, column 3, which is adjacent to 14. Size = 10 pixels, mean = 123/10 = 12.3.
- **Region R2** (seed = 52 at row 1, column 4, accept 44 to 60): the right two columns including 58 and 59 in row 5. Size = 10, mean = 545/10 = 54.5.

- **Region R3** (seed = 60, accept 52 to 68): the pixels 60, 62, 61, 63, 64 in rows 4 and 5. Size = 5, mean = 310/5 = 62.0.

All 25 pixels are assigned (10 + 10 + 5). Note that the means of R2 and R3 (54.5 and 62.0) differ by only 7.5, which is within T = 8. If a region-merging step used the same T, the two regions would be wrongly merged. This is the problem of regions with similar intensity values given in the scenario. It is solved by also checking the boundary strength between R2 and R3 (here a clear 51–58 to 60–64 step) or by using a smaller merge threshold (for example 5).

## 5.3 Edge Linking and Boundary Detection

The edge map of Figure 5.1 shows gaps in the boundaries. Three linking techniques are used.

**(a) Hysteresis thresholding (Canny).** Two thresholds are used. Pixels with gradient above T_high are strong edges. Pixels between T_low and T_high are kept only if they are 8-connected to a strong edge.

**(b) Example:** gradient magnitudes along a boundary are 90, 85, 32, 28, 88, 80, with T_high = 60 and T_low = 25. The values 90, 85, 88 and 80 are strong. The weak values 32 and 28 lie between strong ones, hence they are accepted and the gap is bridged. An isolated weak response of 30 far away from any strong edge is rejected as noise. A single threshold of 60 would leave the boundary broken in two pieces.

**(b) Local processing.** Two edge pixels p and q are linked if they are neighbours and their gradient magnitude and direction are similar: $|M(p) - M(q)| \leq E$ and $|\alpha(p) - \alpha(q)| \leq A$.

**(c) Global processing: Hough transform.** Field boundaries are mostly straight. A line is written as $\rho = x\cos\theta + y\sin\theta$; every edge pixel votes for all $(\rho, \theta)$ cells it can lie on, and a peak gives a complete line even if pixels are missing. **Example:** the edge points (10, 20), (20, 30), (30, 40) all give $\rho = (-x + y)/\sqrt{2} = 10/1.414 = 7.07$ at $\theta = 135°$, so they vote in the same cell and are recognised as one line, and the gaps between them can be filled.

After linking, morphological closing (dilation followed by erosion with a 3 × 3 element) fills one- or two-pixel gaps, and boundary extraction (region minus its erosion, or the morphological gradient) gives a one-pixel-wide closed contour for each region.

## 5.4 Experimental Results

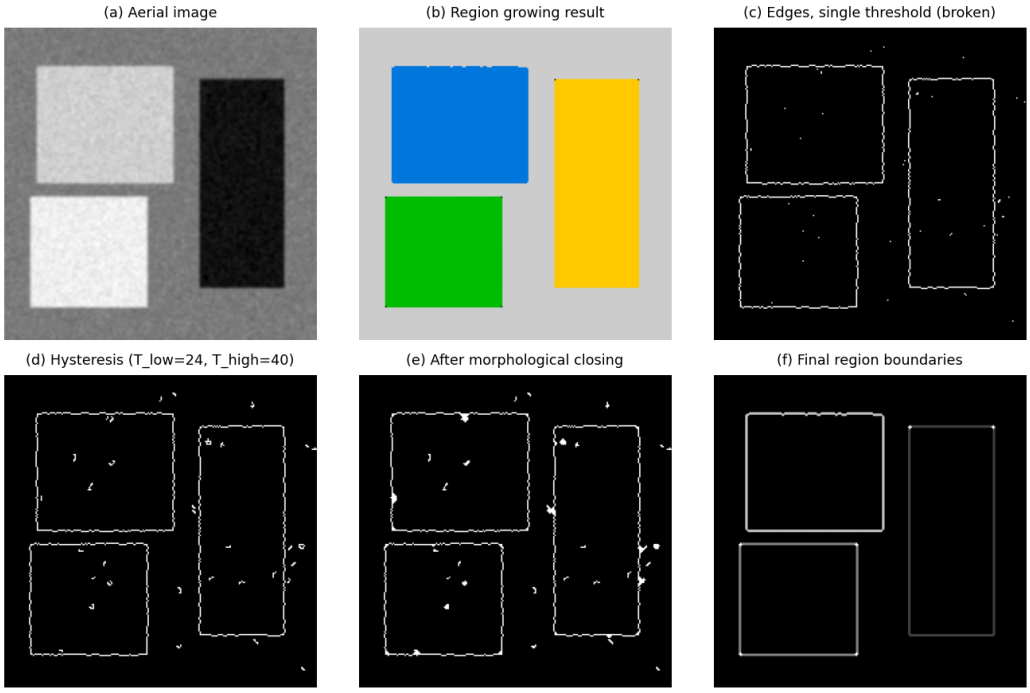



*Figure 5.1: Aerial image, region growing, single-threshold edges, hysteresis edges, closing and final region boundaries*

| Region | True area (pixels) | Region growing area | Error |
|---|---|---|---|
| Field 1 | 9450 | 9424 | 0.3% |
| Field 2 | 7650 | 7646 | 0.05% |
| Water body | 10400 | 10398 | 0.02% |
| Surrounding land | 30100 | 30126 | 0.09% |

*Table 5.1: Area accuracy of region growing (overall pixel accuracy = 99.94%)*

The single-threshold edge map (Figure 5.1c) has 1222 edge pixels with many breaks. Hysteresis thresholding (Figure 5.1d) raised this to 1381 pixels with the weak boundary pieces of the dark water body reconnected, and closing (Figure 5.1e) gave 1514 pixels with the small gaps filled.

## 5.5 How the Techniques Improve Land-Region Identification

- Region growing uses the property that a field is internally homogeneous, so it produces closed regions automatically (overall pixel accuracy 99.94%, with each area within 0.3% of the true area). It does not suffer from gaps in boundaries.
- Edge detection with linking gives the exact position of the boundary, which region growing alone can leak across when two regions have similar

intensity (e.g. R2 and R3 in the worked example). Using the edge map as a barrier that growth cannot cross prevents the leak.

- **Combination:** region growing identifies what each area is (field, water, land) and edge linking decides where the border lies. In practice the boundary strength is used as an additional merge criterion, which improves accuracy for fields with similar crops.
- **Practical benefits:** closed, accurately located boundaries allow the system to compute the area of each field, estimate water-body extent, and track changes over time. Hough-based straight-line boundaries match the geometry of man-made field borders.
- **Limitations:** region growing depends on seed choice and threshold T; textured crops (rows of plants) may fragment a field, which can be corrected by smoothing first or by using texture features.

## 5.6 Conclusion

Region-based segmentation grouped neighbouring pixels of similar intensity into complete regions, while hysteresis edge linking, morphological closing and the Hough transform restored broken boundaries. Combining both gave closed, accurate field, water and land regions (99.94% pixel accuracy in the test), and addresses the two problems of the scenario: similar intensity and discontinuous edges.